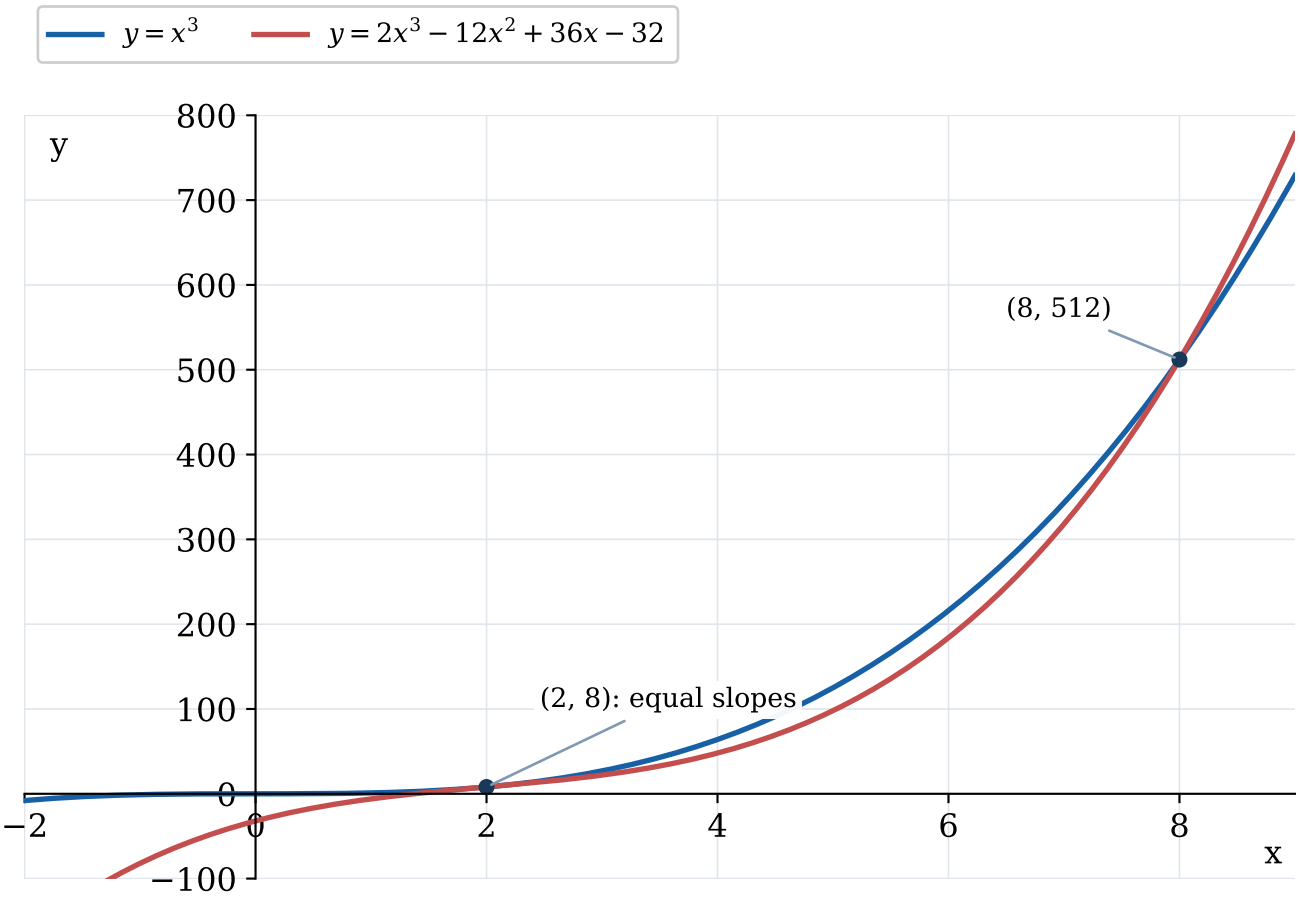
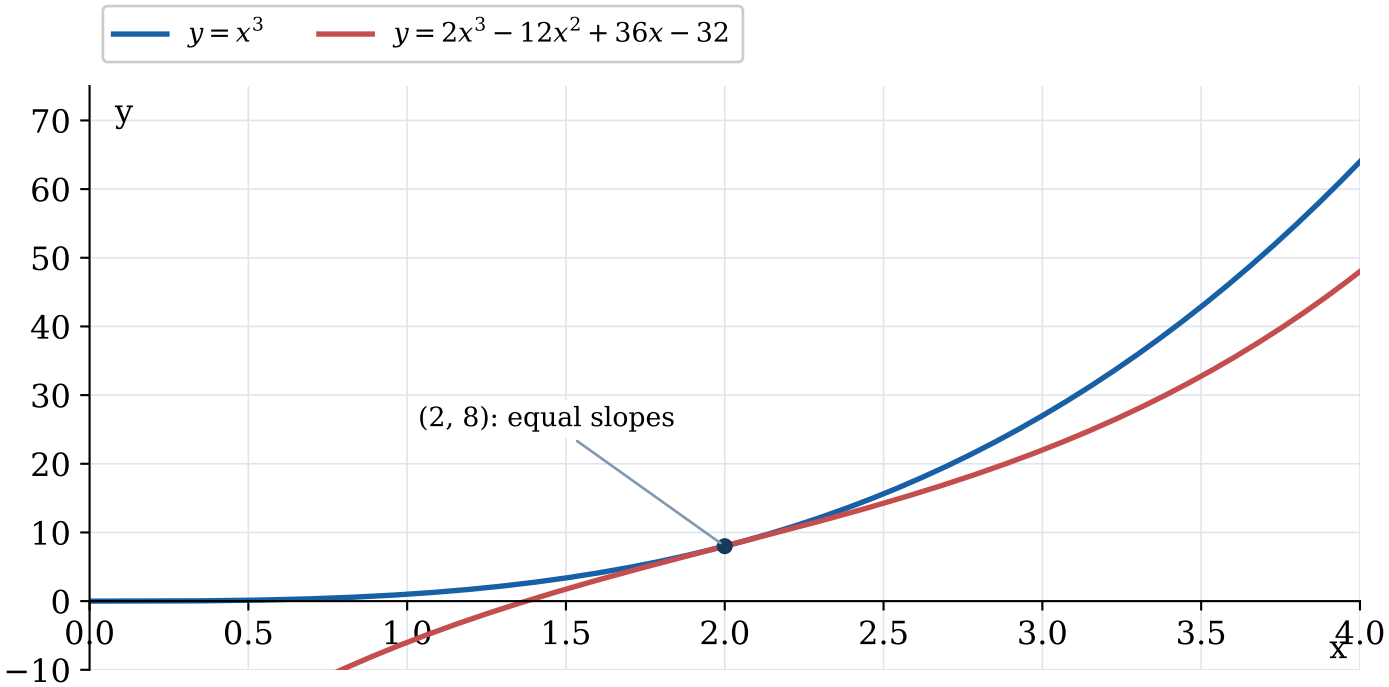
STEP I 2007 Question 8

Case (i): a = 2



Detail around the touching point



The curves touch at (2, 8) and also intersect at (8, 512).
The red curve has derivative $6(x - 2)^2 + 12 > 0$, so it has no turning points.

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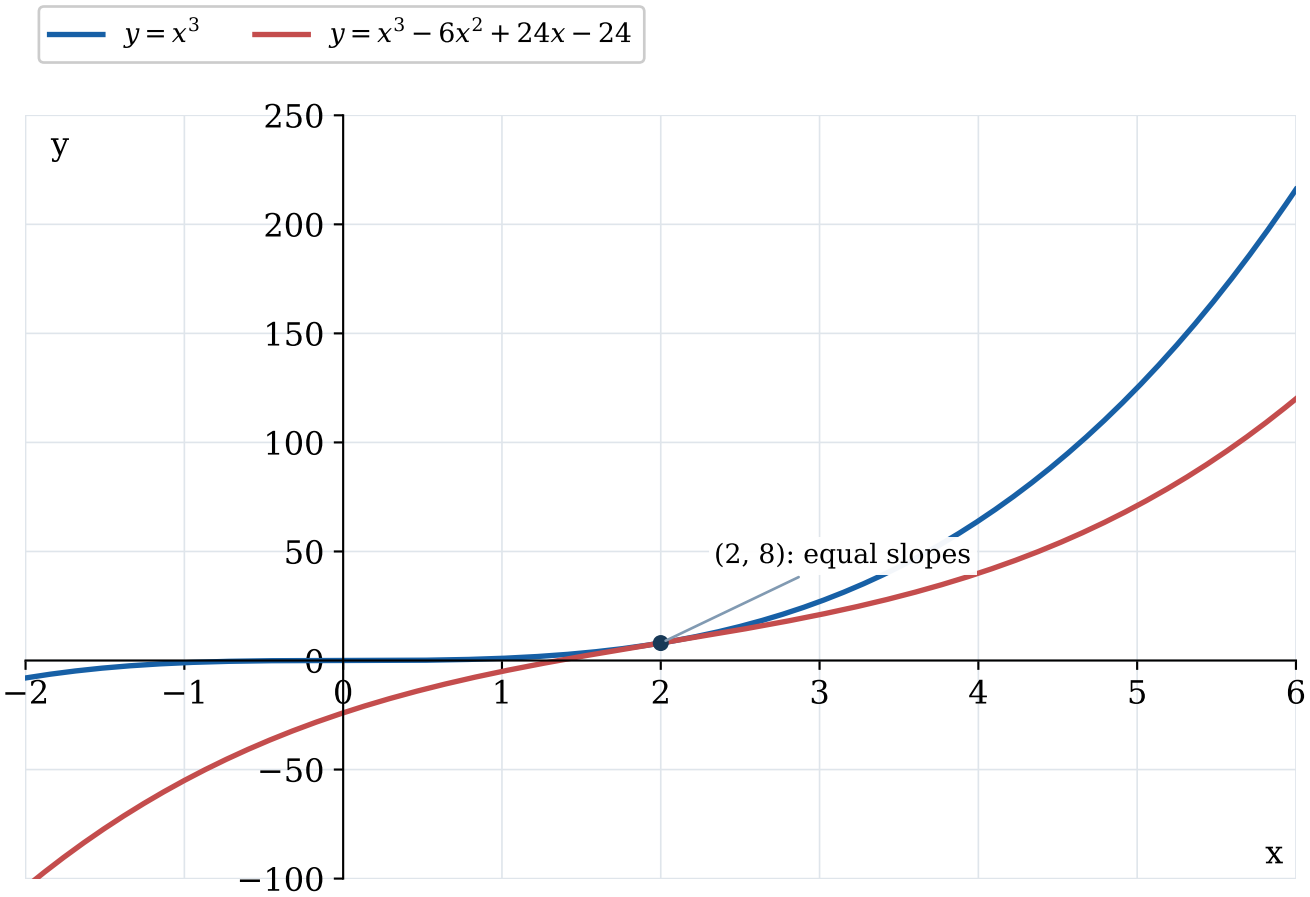
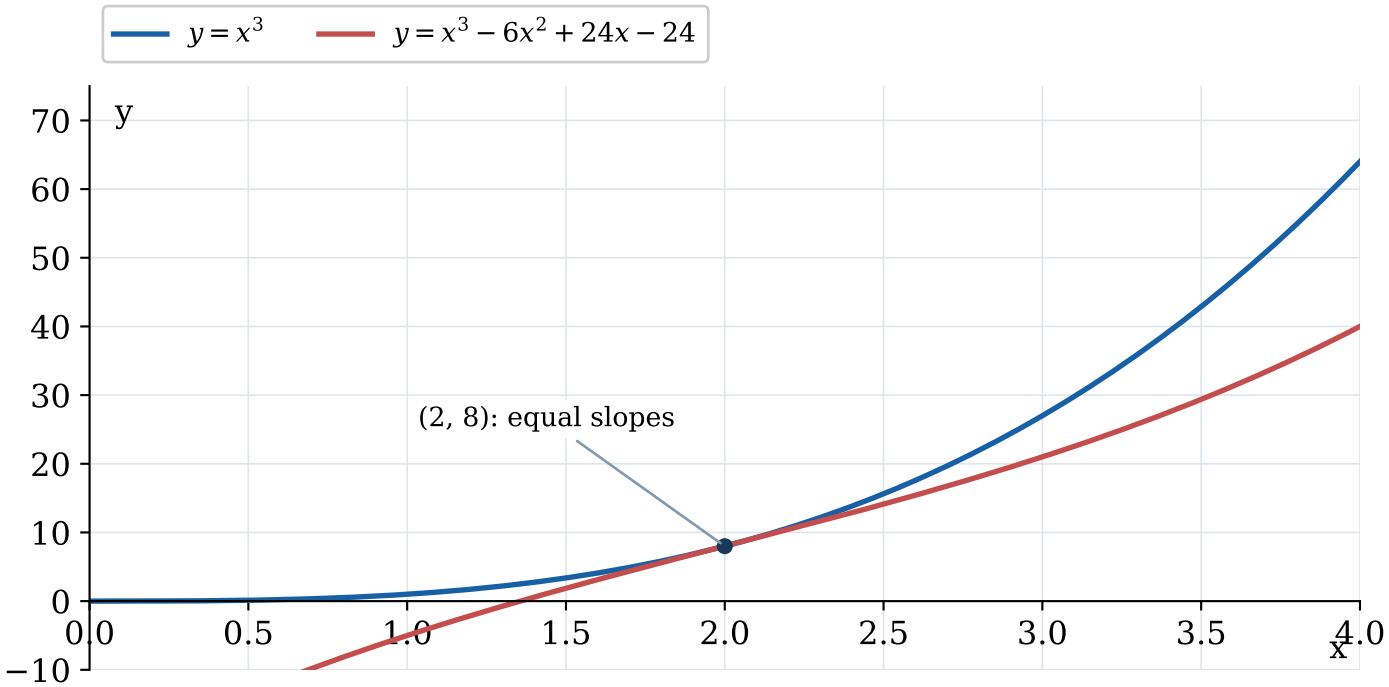
Case (ii): a = 1



Detail around the touching point



The curves touch at (2, 8) and have no other intersection.
The red curve has derivative $3(x - 2)^2 + 12 > 0$, so it has no turning points.

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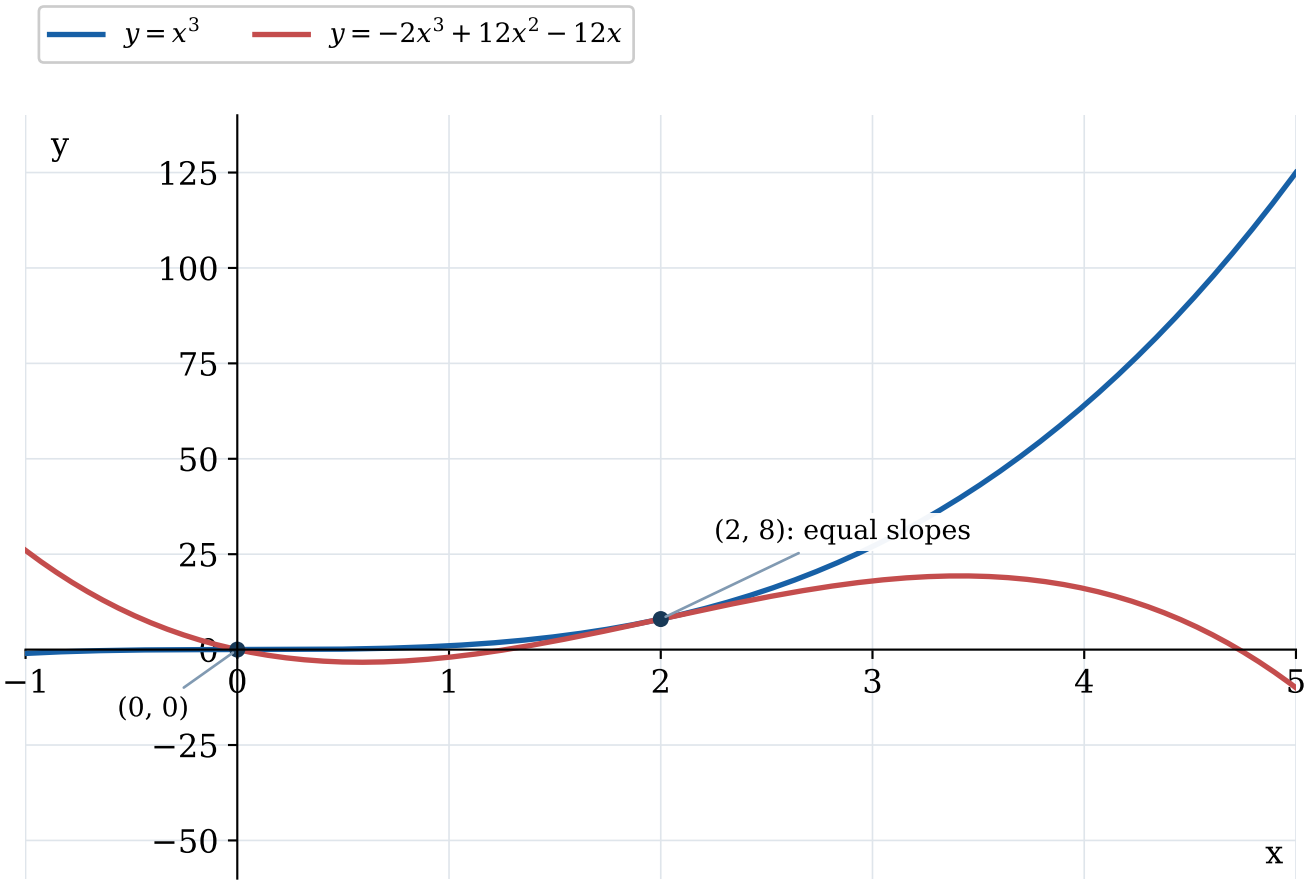
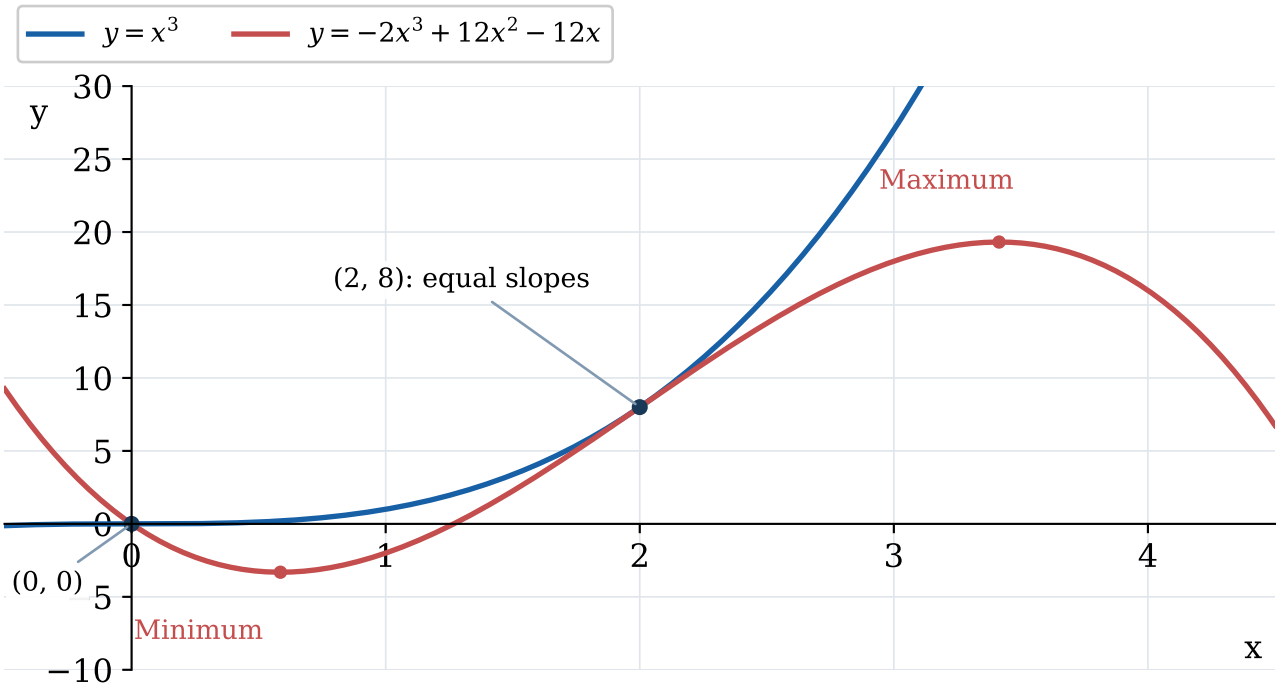
Case (iii): a = -2



Detail showing the two turning points



The curves touch at (2, 8) and also intersect at (0, 0).
The red curve turns at x = 2 − √2 (minimum) and x = 2 + √2 (maximum).